

1. A temperature sensor sensing element have ideal transfer function $R = R_0(1 + T)$, and then the signal condition element have ideal transfer function as $V_{Th} = V_{Th0}(1 + 1.08R)$. With $R_0 = 100 \Omega$ and $V_{Th0} = 1.2 V$. But due to nonlinearity, they have transfer function as $R = R_0(1 + T - 0.2T(T - 1))$ and $V_{Th} = V_{Th0}(1 + 1.08R + 0.003(R - R_0)(R - 2R_0))$. Please plot the output V_{Th} as a function of input T , considering the following four combinations with input $T \in [0.01, 1]$ K:

- A) Ideal sensing element + ideal singal condition element;
- B) Nonlinear sensing element + ideal singal condition element;
- C) Ideal sensing element + nonlinear singal condition element;
- D) Nonlinear sensing element + nonlinear singal condition element.

Please explain why case D) can have better performance than B) and C).

Solution:

- A) Ideal sensing + Ideal signal conditioning:

$$V_A(T) = V_{Th0}[1 + 1.08(R_i(T))]$$

- B) Nonlinear sensing + Ideal signal conditioning:

$$V_B(T) = V_{Th0}[1 + 1.08(R_n(T))]$$

- C) Ideal sensing + Nonlinear signal conditioning:

$$V_C(T) = V_{Th0}[1 + 1.08R_i(T) + 0.003(R_i(T) - R_0)(R_i(T) - 2R_0)]$$

- D) Nonlinear sensing + Nonlinear signal conditioning:

$$V_D(T) = V_{Th0}[1 + 1.08R_n(T) + 0.003(R_n(T) - R_0)(R_n(T) - 2R_0)]$$

1. Analysis of the Sensor's Nonlinearity (Case B):

The nonlinear term in the sensing element is $-0.2T(T - 1)$. For the given input range $T \in (0, 1)$, the term $(T - 1)$ is negative. Therefore, $-0.2T(T - 1)$ yields a **positive value**. This causes the calculated resistance R_n to be higher than the ideal R_i , resulting in a *positive deviation* (upward curve) in the final voltage output.

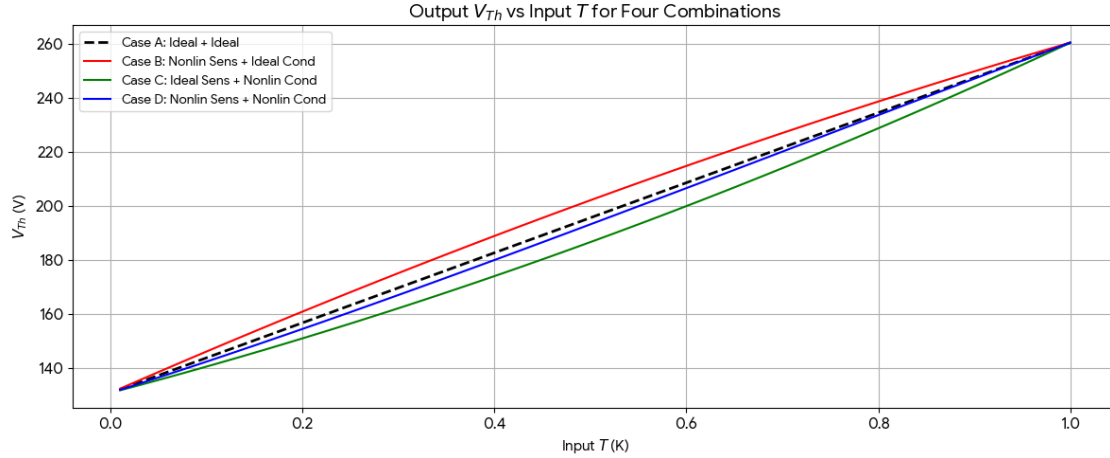
2. Analysis of the Circuit's Nonlinearity (Case C):

The nonlinear term in the signal conditioning element is $0.003(R - R_0)(R - 2R_0)$. Since $T \in [0.01, 1]$, the resistance R varies approximately between 100Ω and 200Ω . Consequently, $(R - R_0) > 0$ and $(R - 2R_0) < 0$. Their product is therefore **negative**. This causes the voltage output to have a *negative deviation* (downward curve) compared to the ideal linear response.

3. The Cancellation Effect (Case D):

When both nonlinear components are connected in series (Case D), the **positive deviation** generated by the sensing element is counteracted by the **negative deviation** generated by the signal conditioning element.

Because these two nonlinearities have opposite trends, they partially cancel each other out. As a result, the total system transfer function in Case D stays much closer to the ideal linear line (Case A) than either Case B or Case C alone.



2. A sensor has transfer function as $O = (K + K_M I_M)I + a + N(I) + K_I I_I = (2 + 0.4I_M)I + 4 + 0.4I_I$. Please design compensation element so that after the compensation element in series, the environment influences are gone.

1. Analysis of the Sensor Model

The given transfer function represents a sensor affected by environmental disturbances:

$$O = (K + K_M I_M)I + a + N(I) + K_I I_I = (2 + 0.4I_M)I + 4 + 0.4I_I$$

By comparing the theoretical model with the numerical equation, we can identify the specific components:

- I : The desired input signal to be measured.
- I_M : The **Modifying Input** (e.g., temperature changes affecting the sensor's gain). Its effect on the sensitivity is $0.4I_M$.
- I_I : The **Interfering Input** (e.g., environmental noise or stray fields). Its effect on the output is $0.4I_I$.
- $a = 4$: The constant zero offset (bias).
- $N(I) = 0$: The nonlinear term is zero in this specific equation.

Objective: Design a compensation element connected in series with the sensor to eliminate the effects of I_M and I_I , ensuring the final output is directly proportional to (or exactly equals) the desired input I .

2. Design of the Compensation Element

To completely remove the environmental influences, the compensation element must perform a two-step mathematical inversion (assuming I_M and I_I can be measured or estimated by reference sensors).

Step 1: Zero Compensation (Offset and Interference Removal)

First, we need to eliminate the additive errors caused by the constant bias and the interfering input. We design a subtraction junction to deduct these terms from the original output O .

- **Operation:** Subtract $(4 + 0.4I_I)$ from O .
- **Intermediate Output (O_1):**

$$O_1 = O - (4 + 0.4I_I) = [(2 + 0.4I_M)I + 4 + 0.4I_I] - (4 + 0.4I_I)$$

$$O_1 = (2 + 0.4I_M)I$$

Step 2: Sensitivity Compensation (Gain Correction)

Next, we need to eliminate the multiplicative error caused by the modifying input, which alters the slope (sensitivity) of the sensor. We design a division block (or a variable gain amplifier) to normalize the signal.

Operation: Divide the intermediate signal O_1 by the environmentally affected sensitivity factor $(2 + 0.4I_M)$.

Final Compensated Output (O_{comp}):

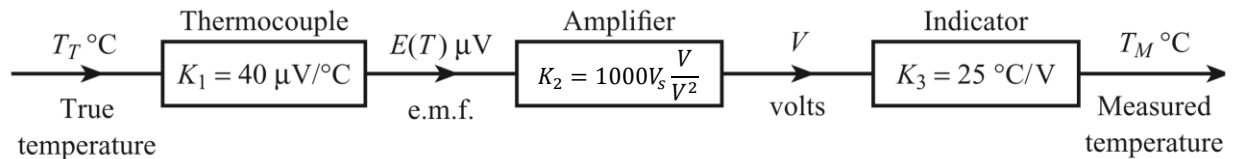
$$O_{comp} = \frac{O_1}{2 + 0.4I_M} = \frac{2 + 0.4I_M}{2 + 0.4I_M} I = I$$

3. Conclusion

The designed compensation element must implement the following mathematical function:

$$O_{comp} = \frac{O - 4 - 0.4I_I}{2 + 0.4I_M}$$

3. The sensor shown below have $K_2 = 1000V_s \frac{V}{V^2}$, i.e., the amplification coefficient is influenced by the voltage supply V_s . Initially, V_s is equal to $1 V^2$. $T_T = 20^\circ C$



A) Please calculate the system error when the $V_s = 1.2 V$.

B) Please add high gain feedback with coefficient of 50 and then add another block with coefficient of 50 after T_M to obtain the final output T'_M , calculate the system error again when the $V_s = 1.0 V$ and $V_s = 1.2 V$.

A) Calculate the system error when $V_s = 1.2 V$

First, we calculate the total forward gain G of the open-loop system. According to the block diagram, the system's transfer function is the product of the individual gains of each module:

$$G = K_1 \cdot K_2 \cdot K_3$$

Given Parameters:

- Thermocouple gain: $K_1 = 40 \mu V/^{\circ}C = 40 \times 10^{-6} V/^{\circ}C$
- Indicator gain: $K_3 = 25^{\circ}C/V$
- True input temperature: $T_T = 20^{\circ}C$

When the voltage supply $V_s = 1.2 V$, the amplifier gain K_2 is:

- $K_2 = 1000 \cdot V_s = 1000 \cdot 1.2 = 1200$

1. Calculate the total open-loop gain (G):

$$G = (40 \times 10^{-6}) \cdot 1200 \cdot 25 = 1.2$$

2. Calculate the measured temperature (T_M):

$$T_M = T_T \cdot G = 20^{\circ}C \cdot 1.2 = 24^{\circ}C$$

3. Calculate the system error:

$$Error = T_M - T_T = 24 - 20 = 4^{\circ}C$$

B) Calculate the system error with feedback when $V_s = 1.0 V$ and $V_s = 1.2 V$

Based on the problem description, the system is modified in two steps:

Adding high gain feedback: A negative feedback loop with a coefficient of $H = 50$ is introduced around the main forward path G . The closed-loop transfer function becomes $\frac{G}{1+GH}$.

Adding a post-processing block: Another block with a coefficient of $K_4 = 50$ is added after T_M to scale the final output T'_M .

The total transfer function of this modified system is:

$$G_{total} = \frac{G}{1 + 50G} \cdot 50$$

And the final measured output is:

$$T'_M = T_T \cdot G_{total}$$

Case 1: When $V_s = 1.0 V$

Under this condition, the amplifier gain $K_2 = 1000$. The forward gain $G = (40 \times 10^{-6}) \cdot 1000 \cdot 25 = 1.0$.

Calculate the final output (T'_M):

$$T'_M = 20 \cdot \left(\frac{1}{1 + 20} 50 \right) = 20 \cdot \frac{50}{51} \approx 19.608^{\circ}C$$

Calculate the system error:

$$Error = 19.608 - 20 = -0.392^{\circ}C$$

Case 2: When $V_s = 1.2 V$

Under this condition, the forward gain $G = 1.2$ (as calculated in Part A).

Calculate the final output (T'_M):

$$T'_M = 20 \cdot \left(\frac{1.2}{1 + 1.2 \cdot 50} 50 \right) = 20 \frac{60}{61} \approx 19.672 \text{ }^\circ\text{C}$$

Calculate the system error:

$$\text{Error} = 19.672 - 20 = -0.328 \text{ }^\circ\text{C}$$

4.

A temperature measurement system consists of a thermocouple, an amplifier, an 8-bit analogue-to-digital converter and a microcontroller with display facilities. Table Prob. 5 gives the model equations and parameters for each element in the system. The temperature of the thermocouple measurement junction is T_1 °C and the temperature of the reference junction is T_2 °C. The microcontroller corrects for T_2 having a non-zero mean value.

- Estimate the mean and standard deviation of the error probability density function when the input temperature T_1 is 100 °C. Treat rectangular distribution as normal with $\sigma = h/\sqrt{3}$.
- Explain briefly what modifications should be made to the system to reduce the quantities calculated in (a).

	Thermocouple	Amplifier	Analogue-to-digital converter	Microcontroller with display
Model equations	$E_{T_1, T_2} = a_0 + a_1(T_1 - T_2) + a_2(T_1^2 - T_2^2)$	$V = K_1 E_{T_1, T_2} + b_1$	$n = K_2 V + b_2$	$T_m = K_3 n + b_3$
Mean values	$\bar{a}_0 = 0$ $\bar{a}_1 = 4.3796 \times 10^{-2}$ $\bar{a}_2 = -1.7963 \times 10^{-5}$ $\bar{T}_2 = 20$	$\bar{K}_1 = 255$ $\bar{b}_1 = 0.0$	$\bar{K}_2 = 0.1$ $\bar{b}_2 = 0.0$ n rounded to nearest integer	$\bar{K}_3 = 1.0$ $\bar{b}_3 = 20$
Statistical distributions	Normal with $\sigma_{a_0} = 0.05$ $\sigma_{T_2} = 2.0$ $\sigma_{a_1} = \sigma_{a_2} = 0.0$	Normal with $\sigma_{b_1} = 5.0$ $\sigma_{K_1} = 0.0$	b_2 has a rectangular distribution of width ± 0.5 $\sigma_{K_2} = 0.0$	$\sigma_{K_3} = 0.0$ $\sigma_{b_3} = 0.0$

Part (a): Statistical Analysis

I. Mean Systematic Error $\bar{E}$

Thermocouple (with reference junction T_2):

$$\bar{E}_{th} = f(T_1, \bar{T}_2) = a_0 + a_1(T_1 - \bar{T}_2) + a_2(T_1^2 - \bar{T}_2^2) \approx \mathbf{3.3312 \text{ mV}}$$

Amplifier:

$$\bar{V} = \bar{K}_1 \bar{E}_{T_1, T_2} + \bar{b}_1 = 255 \times 3.3312 + 0.0 \approx \mathbf{849.46 \text{ mV}}$$

ADC:

$$\bar{n} = \bar{K}_2 \bar{V} + \bar{b}_2 = 0.1 \times 849.46 + 0.0 \approx \mathbf{84.946}$$

Microcontroller:

$$\bar{T}_m = \bar{K}_3 \bar{n} + \bar{b}_3 = 1.0 \times 84.946 + 20 \approx \mathbf{104.95 \text{ }^\circ\text{C}}$$

2. Final Result:

$$\bar{E}_{sys} = \bar{T}_m - T_1 = 104.95 - 100 = \mathbf{+4.95 \text{ }^\circ\text{C}}$$

II. Error Standard Deviation $\sigma_{E_{sys}}$ (Variance Propagation)

1. Core Formulas (Partial Derivatives Squared):

Thermocouple variance:

Amplifier variance:

$$\frac{\partial E}{\partial a_0} = 1$$

$$\frac{\partial E}{\partial T_2} = -\bar{a}_1 - 2\bar{a}_2 \bar{T}_2 = -0.043796 - 2(-1.7963 \times 10^{-5})(20) \approx -0.04308$$

$$\sigma_E^2 = (1)^2 \sigma_{a_0}^2 + \left(\frac{\partial E}{\partial T_2} \right)^2 \sigma_{T_2}^2 = (0.05)^2 + (-0.04308)^2 (2.0)^2 \approx \mathbf{0.00992}$$

Amplifier variance:

$$\sigma_V^2 = (\bar{K}_1)^2 \sigma_E^2 + (1)^2 \sigma_{b_1}^2 = (255)^2 (0.00992) + (5.0)^2 \approx \mathbf{670.18}$$

ADC variance:

$$\sigma_n^2 = (\bar{K}_2)^2 \sigma_V^2 + (1)^2 \sigma_{b_2}^2 = (0.1)^2 (670.18) + \frac{0.5^2}{3} = 6.7018 + 0.0833 \approx \mathbf{6.785}$$

Microcontroller variance:

$$\sigma_{T_m}^2 = (\bar{K}_3)^2 \sigma_n^2 = (1.0)^2 (6.785) = \mathbf{6.785 \text{ (}^\circ\text{C)}^2}$$

2. Final Result:

$$\sigma_{E_{sys}} = \sqrt{\sigma_{T_m}^2} = \sqrt{6.785} \approx \mathbf{2.60 \text{ }^\circ\text{C}}$$

Part (b): Optimization

Cause of the massive +4.95°C Mean Error:

If we combine the nominal models of the ADC and Amplifier, the microcontroller essentially uses a linear inverse equation: $T_m = 25.5E + 20$. However, the thermocouple's actual behavior is a non-linear quadratic curve. This simple linear approximation introduces a severe theoretical error.

Solution (Software): Update the microcontroller's algorithm to use the proper non-linear inverse equation (e.g., polynomial approximation) or a Look-Up Table (LUT) to map the voltage accurately.

Cause of the 2. 60°C Random Fluctuation:

By tracing the variance back, the largest contributor to the final noise ($\sigma_{T_m}^2 = 6.785$) is the variance propagated from the reference junction temperature (T_2) fluctuation, which contributes about 4.82 to the total variance after being amplified.

Solution (Hardware/System): Implement Dynamic Reference Junction Compensation. Instead of assuming T_2 has a constant mean of 20°C, install a dedicated local temperature sensor (like an RTD or thermistor) at the reference junction to measure the true T_2 continuously, and feed this real-time data into the microcontroller to cancel out ambient swings.